Aeroacoustics Project 2

Roi Baruch, 211914866

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Part A

A.1

In the following question, we'll identify the dominant compact source type (monopole/dipole/quadrupole) associated with the noise mechanisms below in low-Mach-number conditions. For a monopole, we'll have a volumetric flow rate Q and no net force, the dipole will have no volumetric flow rate but will have a net force F, and the quadrupole will have neither.

1. Acoustic narrow-band tone at the blade-passing frequency (f_b) from a propeller/rotor due to periodic aerodynamic loading - **dipole** due to the lift force created.

2. Broadband turbulent jet mixing noise from a subsonic single-stream jet - **quadrupole** due to its turbulent behavior and no force or flow rate created.

3. Noise from vortex shedding over a landing-gear strut - **dipole** due to the drag force created.

4. Noise from a cabin outflow/bleed vent (unsteady mass flow through a hole) - **monopole** due to the volumetric flow rate created.

A.2

1. Given two harmonic monopole sources lying on the x_1 axis as follows, with a point on the x_2 axis.

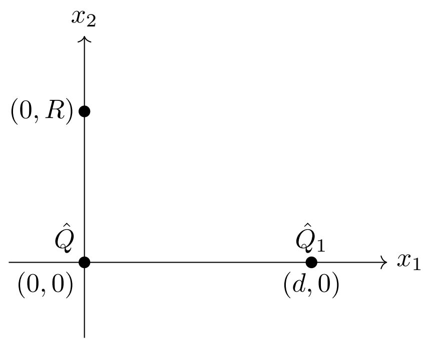

Figure 1: Monopole location on axis.

For a given $\hat{Q}$, we can calculate $\hat{Q}_1$ necessary to produce zero sound pressure at a point on the x_2. Using elementary monopole pressure amplitude:

$$\hat{p}_i = -i\omega\rho_0\hat{Q}_i\frac{e^{ikr_i}}{4\pi r_i} \tag{1}$$

And the principle of superposition, we get that:

$$\hat{p} = \cancel{-i\omega\rho_0}\hat{Q}\frac{e^{ikr_1}}{\cancel{4\pi} r_1}\cancel{-i\omega\rho_0}\hat{Q}_1\frac{e^{ikr_2}}{\cancel{4\pi} r_2} = 0 \tag{2}$$

Substituting in $r_1 = R$ and $r_2 = \sqrt{R^2 + d^2}$ and canceling terms we get:

$$-\hat{Q}\frac{e^{ikR}}{R} - \hat{Q}_1\frac{e^{ik\sqrt{R^2+d^2}}}{\sqrt{R^2+d^2}} = 0 \tag{3}$$

And:

$$\boxed{\hat{Q}_1 = -\frac{\sqrt{R^2+d^2}}{R}e^{ik(R-\sqrt{R^2+d^2})}\hat{Q}} \tag{4}$$

2. Assuming $R \gg d$ and $\lambda \gg d$, the solution simplify to:

$$\hat{Q}_1 = -\frac{\sqrt{R^2}}{R}e^{ik(R-\sqrt{R^2})}\hat{Q} = -\hat{Q} \tag{5}$$

Since kR does not appear in the results, the approximation applies to both the near and far fields.

3. Adding a monopole on the x_1 at $x_1 = -d$:

$$-i\omega\rho_o\hat{Q}\frac{e^{ikR}}{4\pi R} - i\omega\rho_o\hat{Q}_1\frac{e^{ik\sqrt{R^2+d^2}}}{4\pi\sqrt{R^2+d^2}} - i\omega\rho_o\hat{Q}_2\frac{e^{ik\sqrt{R^2+d^2}}}{4\pi\sqrt{R^2+d^2}} = 0 \tag{6}$$

Due to symmetry, the equations at $x_2 = R$ and $x_2 = -R$ are identical. From symmetry we can also say $\hat{Q}_1 = \hat{Q}_2$ Thus, we get:

$$\hat{Q}\frac{e^{ikR}}{4R} + \hat{Q}_1\frac{e^{ik\sqrt{R^2+d^2}}}{2\sqrt{R^2+d^2}} = 0 \tag{7}$$

And $\hat{Q}_1, \hat{Q}_2$ are:

$$\hat{Q}_1 = \hat{Q}_2 = -\frac{\sqrt{R^2 + d^2}}{2R}\,\hat{Q}\,\exp\Big[ik\Big(R - \sqrt{R^2 + d^2}\Big)\Big] \tag{8}$$

A.3

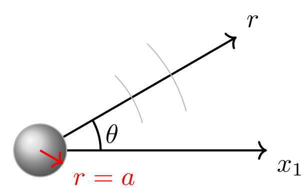

Figure 2: Translating sphere oscillating in the x_1 direction.

Considering a sphere of radius a that translates back and forth in the x_1 direction with velocity v_1, pulsating with ω. We'll ignore any Viscous effects, thus only the radial velocity of the surface of the sphere is considered. The boundary conditions for the sphere are:

$$\text{BC}: \quad \begin{cases} v_r = v_0 \cos\theta\, e^{-i\omega t}, \quad \hat{v}_r = v_0 \cos\theta, r = a, \\ \nabla^2 \hat{p} + k^2 \hat{p} = 0, \qquad\qquad\qquad r \geq a \end{cases} \tag{9}$$

The solution to the wave equation is:

$$\hat{p}(r) = \frac{\hat{A}e^{ikr}}{r} \tag{10}$$

Using the fact that any derivative of a solution to the wave equation is also a solution to the wave equation, we can find a solution that also satisfies the BC:

$$\hat{p}(r) = \frac{\partial}{\partial x_1}\frac{\hat{A}e^{ikr}}{r} \tag{11}$$

Using the chain rule:

$$\hat{p}(r) = \frac{\partial r}{\partial x_1}\frac{\partial}{\partial r}\frac{\hat{A}e^{ikr}}{r} = ik\cos\theta\,\hat{A}\,\frac{e^{ikr}}{r}\left(1 - \frac{1}{ikr}\right) \tag{12}$$

From the acoustic momentum conservation, we obtain:

$$\hat{\mathbf{v}} \cdot \mathbf{n}_r = \frac{1}{i\omega\rho_0}\frac{\partial \hat{p}}{\partial r} \tag{13}$$

The BC at the sphere surface states:

$$= \frac{\partial p_0}{\partial r} = \frac{ik\cos\theta}{i\omega\rho_0}\,\hat{A}\,\frac{e^{ikr}}{r}\left(ik - \frac{2}{r} + \frac{2}{ikr^2}\right) \tag{14}$$

Thus, substituting the BC into the momentum conservation yields

$$\hat{\mathbf{v}} \cdot \mathbf{n}\big|_{r=a} = \frac{1}{i\omega\rho_0} \frac{\partial \hat{p}}{\partial r}\bigg|_{r=a} = v_0 \cos\theta \tag{15}$$

And we can solve for A:

$$v_0 \cos\theta = \frac{\cos\theta}{c_0\rho_0} \left(\frac{\hat{A}e^{ika}}{a}\right) \left(ik - \frac{2}{a} + \frac{2}{ika^2}\right) \tag{16}$$

Hence:

$$\hat{A} = ika^3 c_0\rho_0 v_0 \left(\frac{e^{-ika}}{2 - (ka)^2 - 2ika}\right) \tag{17}$$

Substituting A, we get:

$$\hat{p}(r) = ik\cos\theta ika^3\, \rho_0 c_0\, v_0\, \underbrace{\left(\frac{e^{-ika}}{2 - (ka)^2 - 2ika}\right)}_{non\,compact}\, \underbrace{\frac{e^{ikr}}{r}}_{far\,field}\, \underbrace{\left(1 - \frac{1}{ikr}\right)}_{near\,field} \tag{18}$$

Approximating this result for a source $ka << 1$:

$$\hat{p}(r) = \frac{ika\cos\theta}{2} \left(\frac{i\omega\rho_0 v_0 a^2\, e^{ikr}}{r}\right) \left(1 - \frac{1}{ikr}\right) \tag{19}$$

And for the far field $kr >> 1$:

$$\hat{p}(r) = ik^3a^3\rho_0 c_0\, v_0 \cos\theta e^{ikr} \left(\frac{e^{-ika}}{2 - (ka)^2 - 2ika}\right) \frac{i}{kr} \tag{20}$$

Using this factorization to plot the error in dB in the compact approximation, as a function of ka, and in the far-field approximation as a function of kr with the error in dB being $20\log_{10}\left(\Re\left\{\frac{\text{approx}}{\text{exact}}\right\}\right)$

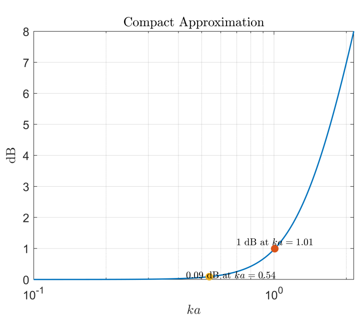

Figure 3: Compact approximation error

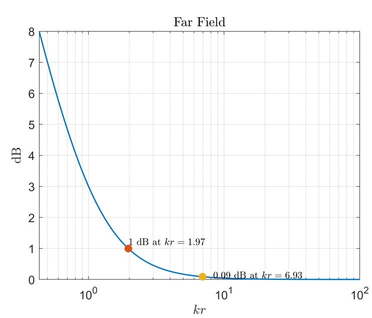

Figure 4: Far-field approximation error

Figure 5: Error for various dipole approximations

A.4

For an aircraft approaching to land, with its engine operating at idle, that generates tonal noise that is recorded by a far-field observer located on the ground at a radial distance of $r = 90$ m from the aircraft. The vehicle emits two narrow-band tonal components: the first has a sound pressure level of 63 dB (re 20 μPa) at a frequency $f_1 = 500$ Hz, and the second has a sound pressure level of 59 dB (re 20 μPa) at a frequency $f_2 = 2000$ Hz.

Using the standard speed of sound of $c_0 = 343$ m/s, the air density is $\rho_0 = 1.225$ kg/m^3, and the atmospheric pressure is $p_0 = 101{,}325$ Pa, and neglecting the Doppler effects.

1. The SPL of the overall sound heard by the observer can be calculated using the definition of SPL:

$$SPL_{tot} = 10\log_{10}\left(\frac{p^2_{rms,tot}}{p^2_{ref}}\right) \tag{21}$$

With $p^2_{rms,tot} = p^2_{1,rms} + p^2_{2,rms}$ as proved in the last assignment. We can obtain each SPL with the inverse relation:

$$p^2_{rms,i} = 10^{\frac{SPL_i}{10}} \cdot p^2_{ref} \tag{22}$$

Thus, we get:

$$p^2_{rms,1} = 7.981 \times 10^{-4}\ [Pa^2],\ p^2_{rms,2} = 3.177 \times 10^{-4}\ [Pa^2] \tag{23}$$

And the total SPL is:

$$\boxed{SPL_{tot} = 64.45 \text{ dB (re 20 } \mu\text{Pa)}} \tag{24}$$

2. To determine the required size of the noise-generating device,consider itd as a compact source at 500 Hz and at 2000 Hz, we'll start with the definition of a compact source as $kd \ll 1$ or:

$$d \ll \frac{1}{k} \rightarrow d \ll \lambda = \frac{c_0}{\omega} \tag{25}$$

for a compact source at $f = 500Hz$ we get:

$$d \ll 0.1091\,[m] \tag{26}$$

and for a compact source at $f = 500Hz$ we get:

$$d \ll 0.02730\,[m] \tag{27}$$

3. Estimating the volume velocity amplitude $\hat{Q}$ of each of the two sources f_1 , f_2,we can use the equation of the monopole

$$\hat{Q} = \frac{-4\pi r e^{-ikr}}{i\omega\rho_0}\hat{p} \tag{28}$$

And the absolute value is:

$$|\hat{Q}| = \frac{2r}{f\rho_0}|\hat{p}| \tag{29}$$

Now ,for the case of an harmonic wave we know $\hat{p}$:

$$\hat{p} = \sqrt{2}p_{rms} \tag{30}$$

Thus we get:

$$|\hat{Q}_i| = \frac{\sqrt{8}r}{f\rho_0}p_{rms,i} \tag{31}$$

$$Q_1 = 1.174 \times 10^{-2} \;\left[\frac{m^3}{s}\right] \tag{32}$$

$$Q_2 = 1.852 \times 10^{-2} \;\left[\frac{m^3}{s}\right] \tag{33}$$

4. For incompressible flow in the pipes, we can use Bernoulli's equation to estimate the velocity:

$$\frac{1}{2}\rho_0 u^2 = \Delta p \tag{34}$$

The velocity is:

$$\sqrt{\frac{2\Delta p}{\rho_0}} = 114.3 \;\left[\frac{m}{s}\right] \tag{35}$$

From this we can estimate the diameter through:

$$Q = uS = \frac{\pi D^2}{4}u \tag{36}$$

$$D = \sqrt{\frac{4Q}{\pi u}} \tag{37}$$

And we can calculate each diameter:

$$D_1 = 0.0114 \,[m] \tag{38}$$

$$D_2 = 0.0185 \,[m] \tag{39}$$

Part B

B.1

1. Considering the following retarded time equation:

$$c_0\,(t-\tau) = |\mathbf{x} - \mathbf{y} - \mathbf{v}_0\,(t-\tau)| \tag{40}$$

We take this equation for the case of consider a point source located at $y = 0$ and emitting a pulse at $\tau = 0$:

$$c_0\,(t) = |\mathbf{x} - \mathbf{v}_0 t| \tag{41}$$

Now using the definition of the absolute value:

$$c_0\,(t) = \sqrt{(\mathbf{x} - \mathbf{v}_0 t)^2} = \sqrt{|\mathbf{x}|^2 - 2(\mathbf{x} \cdot \mathbf{v}_0)t + \mathbf{v}_0^2 t^2} \tag{42}$$

Rearranging we can get

$$\boxed{\left(c_0^2 - |\mathbf{v}_0|^2\right) t^2 + 2\,(\mathbf{x} \cdot \mathbf{v}_0)\, t - |\mathbf{x}|^2 = 0} \tag{43}$$

2. This is quadratic equation, with the coefficients being:

$$\begin{aligned} a &= c_0^2 - |\mathbf{v}_0|^2 \\ b &= 2\,(\mathbf{x} \cdot \mathbf{v}_0) \\ c &= -\,|\mathbf{x}|^2 \end{aligned} \tag{44}$$

Thus the condition for a real solution for t is:

$$\Delta = b^2 - 4ac > 0 \tag{45}$$

And for this case:

$$4(\mathbf{x} \cdot \mathbf{v}_0)^2 + 4|\mathbf{x}|^2(c_0^2 - |\mathbf{v}_0|^2) > 0 \tag{46}$$

Using $(\mathbf{x} \cdot \mathbf{v}_0) = |\mathbf{x}||\mathbf{v}|\cos(\theta)$, multiplying and dividing the second term by $|\mathbf{v}|^2$:

$$4|\mathbf{x}|^2|\mathbf{v}|^2\cos^2(\theta) + 4|\mathbf{x}|^2|\mathbf{v}|^2(\frac{c_0^2}{|\mathbf{v}|^2} - 1) > 0 \tag{47}$$

Or

$$4|\mathbf{x}|^2|\mathbf{v}|^2(\frac{c_0^2}{|\mathbf{v}|^2} + cos^2(\theta) - 1) > 0 \tag{48}$$

As $4|\mathbf{x}|^2|\mathbf{v}|^2$ is positive, the identity $\sin^2(\theta) + \cos^2(\theta) = 1$, and from the definition of the Mach number, we are left with:

$$\frac{1}{M^2} - \sin^2(\theta) > 0 \tag{49}$$

And we get that a real solution can only exist in the Mach cone as defined:

$$\boxed{\sin\theta < \frac{1}{M}} \tag{50}$$

B.2

1. For the given case of A source moving with constant speed and a subsonic Mach number M along the x_1-direction in a two-dimensional Cartesian coordinate system (x_1, x_2), with a stationary observer located on the ground directly beneath the source, with d denoting the minimum distance between the source and the observer. The fluid medium is at rest ($\mathbf{v}_0 = 0$). We can calculate the frequency heard by the observer.

$$\frac{\omega_x}{\omega_y} = \frac{1}{1 - M_y \cos(\theta)} \tag{51}$$

$\cos(\theta)$ can be obtained from the simple geometry:

$$\cos(\theta) = -\frac{v_s \tau}{c_0(t-\tau)} \tag{52}$$

And using Pythagoras's theorem we can find $c_0(t-\tau)$:

$$c_0(t-\tau) = \sqrt{(v_s^2 \tau^2 + d^2)} = d\sqrt{1 + \frac{M^2 c_0^2 \tau^2}{d^2}} \tag{53}$$

Now normalizing the time scales using $\tilde{t} = \frac{c_0 t}{d}$ we can rewrite $c_0(t-\tau)$ as:

$$c_0(t-\tau) = d\sqrt{1 + M^2 \tilde{\tau}^2} \tag{54}$$

Thus, $\cos\theta$ is:

$$\cos\theta = -\frac{M\tilde{\tau}}{\sqrt{1 + M^2 \tilde{\tau}^2}} \tag{55}$$

And for the case of $r = d$, corresponding to $\tilde{\tau} = 0$ and $\tilde{t} = 1$, n_r and v_s are perpendicular therefore Doppler has no effect and $\omega_0 = \omega_y$. Now we can show the normalized frequency as a function of the normalized time, and plot for various Mach numbers:

$$\frac{\omega_x}{\omega_y} = \frac{1}{1 + \frac{M^2 \tilde{\tau}}{\sqrt{1 + M^2 \tilde{\tau}^2}}} \tag{56}$$

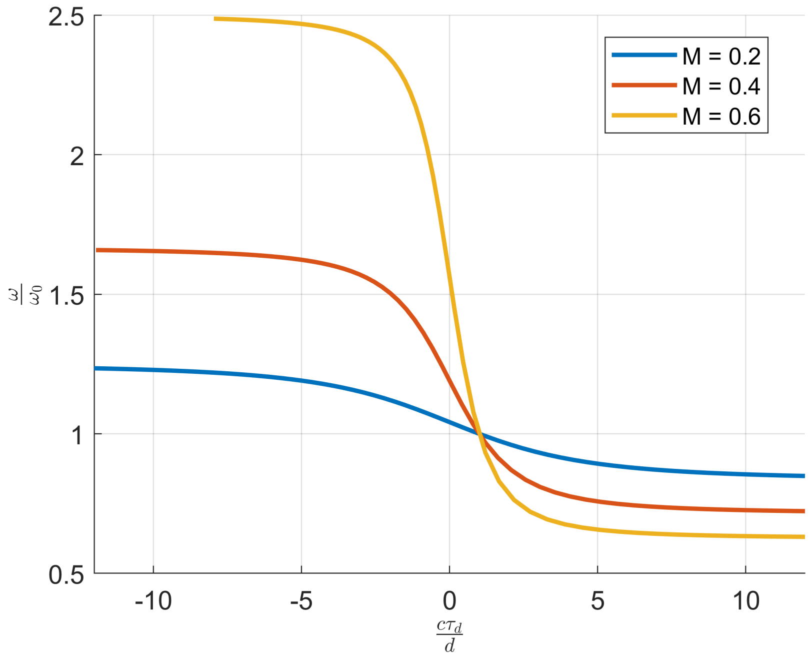

Figure 6: Frequency change due to the Doppler effect

2. In the case that both the source and the observer were kept stationary in a fluid medium that is characterized by a velocity vector of $\mathbf{v}_0 = (v_0, 0)$, both observer and source still not have any relative velocity, thus no Doppler effect occurs, and $\dfrac{\omega_x}{\omega_0} = 1$.

B.3

1. For the given free-field Green's function:

$$G_0(\mathbf{x}, t \mid \mathbf{y}, \tau) = \frac{\delta(t - \tau - |\mathbf{x} - \mathbf{y}|/c_0)}{4\pi|\mathbf{x} - \mathbf{y}|}, \tag{57}$$

With a fixed observer location $\mathbf{x}$ and a fixed time t, along with $\mathbf{y}$ and τ for the source location and emission time. We can look at the properties of the Green's function firstly at $t > \tau$, the Dirac delta is zero expect for:

$$t - \tau = \frac{|\mathbf{x} - \mathbf{y}|}{c_0}. \tag{58}$$

Thus the time delay between the source and the observe represented by $t - \tau$ is equal to the time required for a sound wave to propagate from the source to the observe. For the case of a fixed observer at a fixed time, the equation defines a spherical surface of possible source locations. As τ increases, this sphere shrinks until it collapses to a single point when the wave reaches the observer. Now for the case that $t < \tau$, the Dirac delta can never be equal to zero, thus the Green's function is zero $G_0(\mathbf{x}, t \mid \mathbf{y}, \tau) = 0$. This result shows the causality condition, as the observer cannot receive a sound before it is produced.

2. Now for the opposite case of a fixed $\mathbf{y}$ and τ, the same causality argument still holds for $t < \tau$. For $t > \tau$ the Green's function now represents a spherical surface of all possible observers, getting bigger as t grows and showing the outgoing impulsive wave. Showing that Green's function satisfies the wave equation, away from the source $\mathbf{x} \neq \mathbf{y}$ and:

$$\frac{1}{c_0^2}\frac{\partial^2 G}{\partial t^2} - \frac{\partial^2 G}{\partial y_i^2} = 0 \tag{59}$$

Transferring to a spherical coordinate system and using the given identity:

$$\frac{1}{c_0^2}\frac{\partial^2 G_0}{\partial t^2} - \left(\frac{\partial^2 G_0}{\partial r^2} + \frac{2}{r}\frac{\partial G_0}{\partial r}\right) = 0 \tag{60}$$

Applying the chain rule on t:

$$\frac{\partial G_0}{\partial t} = \frac{\delta'(\alpha)}{4\pi r}, \quad \frac{\partial^2 G_0}{\partial t^2} = \frac{\delta''(\alpha)}{4\pi r}, \quad \frac{1}{c_0^2}\frac{\partial^2 G_0}{\partial t^2} = \frac{\delta''(\alpha)}{4\pi r c_0^2}. \tag{61}$$

Now applying the chain rule on r:

$$\frac{\partial G_0}{\partial r} = \frac{1}{4\pi}\left(-\frac{\delta'(\alpha)}{rc_0} - \frac{\delta(\alpha)}{r^2}\right), \quad \frac{\partial^2 G_0}{\partial r^2} = \frac{1}{4\pi}\left(\frac{\delta''(\alpha)}{rc_0^2} + \frac{2\delta'(\alpha)}{r^2 c_0} + \frac{2\delta(\alpha)}{r^3}\right). \tag{62}$$

Using the terms we can represent the Laplace operator:

$$\nabla^2 G_0 = \frac{1}{4\pi}\frac{\delta''(\alpha)}{rc_0^2}. \tag{63}$$

Using this Laplace operator in the wave equation:

$$\frac{1}{c_0^2}\frac{\partial^2 G_0}{\partial t^2} - \nabla^2 G_0 = \frac{\delta''(\alpha)}{4\pi r c_0^2} - \frac{\delta''(\alpha)}{4\pi r c_0^2} = 0 \tag{64}$$

for $(x, t) \neq (y, \tau)$, satisfying the wave equation away from the source. Now, for a source surface, we saw in the lectures that Green's function allows for the application of inhomogeneous boundary conditions, resulting in the pressure perturbations:

$$p'(x,t) = \int_{-T}^{T}\int_S \left(p'(y,\tau)\frac{\partial G}{\partial y_i} - G\frac{\partial p'(y,\tau)}{\partial y_i}\right) n_i \, dS(y)\, d\tau \tag{65}$$

Part C

C.1

Yes. The self-sustained hydro-acoustic feedback loop can take place in a flow entering the cavity that is characterized by a laminar boundary layer, as long as the boundary layer is compressible, as acoustic propagation is a compressible phenomenon. The shear-layer dynamics and vortex stretching play a vital role in the formation of the feedback loop, with vorticity being the key flow parameter.

C.2

1. As $f = \frac{U}{\lambda}$, we can use the following parameters to write an expression for the hydrodynamic vortex shedding frequency f_v and the acoustic frequency f_a:

$$f_v = \frac{U_v}{\lambda_v} = \frac{kU_\infty}{\lambda_v} \tag{66}$$

   And:

$$f_a = \frac{U_a}{\lambda_a} \tag{67}$$

   In the case of self-sustained hydro-acoustic oscillation in a cavity, the two frequencies must be equal ($f = f_a = f_v$).

2. The basic kinematic expression is $\Delta x = v\Delta t$, now for the vortices, the velocity is u_v and the length scale is $n_v\lambda_v - (L - x_v)$ all the wave starting from a location x_v, thus for a time interval of Δt_0 we have:

$$n_v\lambda_v - (L + x_v) = U_v t_0 \tag{68}$$

   The same equation can be written for the acoustic waves:

$$L - n_a\lambda_a = U_a t_0 \tag{69}$$

3. Dividing the two equations:

$$\frac{n_v\lambda_v - (L + x_v)}{L - n_a\lambda_a} = \frac{U_v\cancel{t_0}}{U_a\cancel{t_0}} \tag{70}$$

$$\frac{U_v}{U_a}(L - n_a\lambda_a) = n_v\lambda_v - (L + x_v) \tag{71}$$

$$L\left(1 + \frac{U_v}{U_a}\right) = \frac{U_v}{U_a}n_a\lambda_a + n_v\lambda_v - x_v \tag{72}$$

   And now for $x_v = \alpha\lambda_v$:

$$L\left(1 + \frac{U_v}{U_a}\right) = \frac{U_v}{U_a}n_a\lambda_a + \lambda_v(n_v - \alpha_v) \tag{73}$$

4. For the case of the self-sustained hydro-acoustic oscillations ($f_a = f_v$), using the definition of $f = U/\lambda$:

$$L\left(1+\frac{U_v}{U_a}\right) = \frac{U_v}{f}n_a + \frac{U_v}{f}\lambda_v(n_v - \alpha_v) \tag{74}$$

As $U_v = kU_\infty$ we can extract f:

$$f = \frac{kU_\infty(n_a + n_\nu - \alpha_\nu)}{L\left(1+\frac{kU_\infty}{U_a}\right)} \tag{75}$$

For the sound wave traveling at the speed of sound $U_a = c_0$ and $n = n_a + n_v$ we get the Rossiter's semi-empirical model:

$$f = \frac{U_\infty}{L}\frac{n - \alpha_\nu}{\left(\frac{1}{k} + M_\infty\right)} \tag{76}$$

With α_ν - phase difference between vortex shedding and acoustic radiation, and k the ratio between free stream and shear layer velocities.

C.3

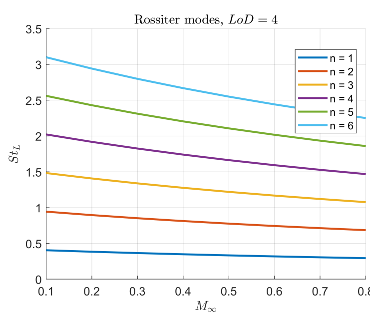

Figure 7: St_L as a function of M_∞ for the first six Rossiter modes.

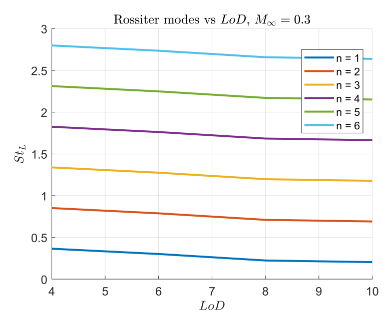

Figure 8: St_L as a function of LoD for the first six Rossiter modes.

Figure 9: Effects of Mach or LoD on the Strouhal number from the first six Rossiter modes.

From the plots we can see that increasing both the mach number and LoD ratio results in the frequency of the mode decreasing. For a constant mach number or LoD ratio, the higher the mode we get a higher frequency.

C.4

Considering an aircraft that approaches to land along a straight path y_1 parallel to the ground at an altitude of $h = 500$ ft and a constant airspeed of $M_\infty =$

0.2 (TAS). During this approach, a total of three landing gears are deployed (one at the nose and two main landing gears close to the wing root), each from its rectangular bay (cavity). Assume each landing gear consists of wheels of diameter 0.52 m and a strut assembly of diameter 0.18 m, and that each cavity has dimensions $L = 1$ m and $D = 0.5$ m. The principal source of noise from the landing gear elements is the unsteady side force resulting from vortex shedding, for which the fundamental frequency satisfies $St \approx 0.2$, where U_∞ is the freestream velocity and d is the characteristic component diameter. Assume the speed of sound is $c_0 = 343$ m/s. In addition, each bay element exhibits four dominant Rossiter modes. Neglecting any aerodynamic or acoustic interactions between the landing gears and bays, and assuming that the landing gears and bays are subjected to the same freestream velocity.

1. We can determine if the wheels, strut assembly and bays can be considered acoustically compact sources, and the relevant Rossiter modes satisfy the compactness criteria. Using $St = 0.2 = \frac{fd}{U_\infty}$, we can extract each frequency, giving us the wavenumber and ka. Additionally we can useRossiter'ss semi-empirical model:

Noise source	Strut	Wheel	Mode 1	Mode 2	Mode 3	Mode 4
f [Hz]	76.2	26.4	26.3	61.4	96.53	132
ka	0.25	0.25	0.48	1.12	1.77	2.41

   Thus, we get that only the strut,wheel, and the first Rossiter mode satisfy $ka \ll 1$ and can be considered as compact.

2. The shortest distance d to the microphone is:

$$d = \frac{h}{\sin(45)} = 215.5 \ [m] \tag{77}$$

   Calculating kd for $r = d$ and checking if $kr \gg 1$:

Noise source	Strut	Wheel	Mode 1	Mode 2	Mode 3	Mode 4
ka	301	104	104	242	381	520

   Thus, all the sources and modes satisfy $kr \gg 1$ and are in the far field.

3. The first harmonic include both the wheel and the cavity's first mode at $f = 26.4\,[Hz]$ amplifying each other. The second one is the strut at $f = 76.2\,[Hz]$, as all the other sources aren't compact. Now taking into account the Doppler effect with $-1 \leq \cos\theta \leq 1$ we get:

$$f = 26.3\,[Hz] \rightarrow f_{mic} \in [21.98, 32.98] \tag{78}$$

$$f = 76.22\,[Hz] \rightarrow f_{mic} \in [63.52, 95.28] \tag{79}$$